Calc I lecture 14: Linear approximations and differentials

We now revisit the concept of a tangent line.

Recall: if $f(x)$ is differentiable at $x=a$, then

the tangent line to the graph $y=f(x)$ at $x=a$ is given by

$$\boxed{y = f(a) + f'(a)\cdot(x-a)}$$

this comes from point-slope formula

Point: $(a, f(a))$

slope: $f'(a)$

Key observation: if we "zoom in" on the graph near $x=a$, the tangent line starts looking very similar to the original function $f(x)$:

$$\boxed{f(x) \approx f(a) + f'(a)\cdot(x-a)}$$

at least if x is close to a.

"approximately equals"

This approximation is called the linear approximation of $f(x)$ at $x=a$

Note: the tangent line defines a function, which we denote

$$\boxed{L(x) = f(a) + f'(a)\cdot(x-a)}$$

$L(x)$ is called the linearization of $f(x)$ at $x=a$

Note: $L(x)$ depends on the choice of a.

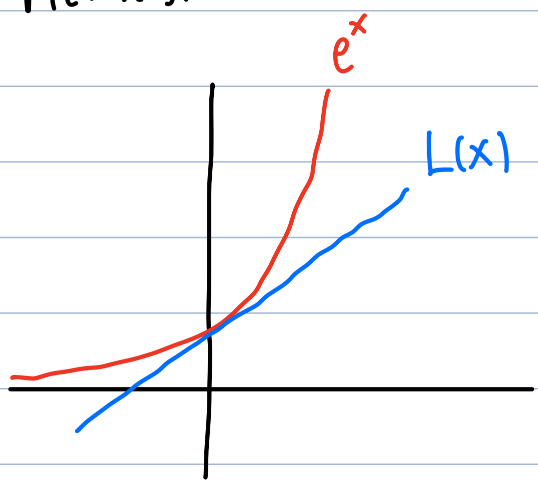
Example: Find the linear approximation of $f(x) = e^x$ at $a=0$.

$$L(x) = f(a) + f'(a)\cdot(x-a)$$

$$L(x) = f(0) + f'(0)\cdot(x-0) \Rightarrow \boxed{L(x) = 1 + 1\cdot x = 1+x}$$

$f(0) = e^0 = 1$ $\quad f'(x) = e^x \Rightarrow f'(0) = e^0 = 1$

Pictures:



Example: use this to approximate $e^{0.01}$

$e^x \approx L(x) = 1+x$ (when x is close to 0)

plug in $x = 0.01$ $\Rightarrow$ $\boxed{e^{0.01} \approx 1+0.01 = 1.01}$

Note: the closer x is to a, the better the linear approximation will be (generally speaking)

Example a) Find the linearization of $f(x) = \ln(x+1)$ at $a=0$

b) use this to approximate $\ln(0.98)$

a) $L(x) = f(a) + f'(a)(x-a)$

$a=0$: $L(x) = f(0) + f'(0)(x-0)$

$f(0) = \ln(0+1) = \ln(1) = 0$

$f(x) = \ln(x+1) \Rightarrow f'(x) = \frac{1}{x+1} \cdot 1 \Rightarrow f'(0) = \frac{1}{0+1} = 1$

$L(x) = 0 + 1 \cdot x = x$

So: $\boxed{\ln(x+1) \approx x}$ when x is close to 0.

b) $\boxed{\ln(0.98) = \ln(-0.02+1) \approx -0.02}$

by a)

This simple idea can be used to estimate function values at points that are "nearby" already-known points.

<u>Example</u>: Use linear approximations to estimate the following quantities:

Idea: write the given quantity as $f(x)$ for some function and use linear approx. at nearby point a

$f(x) \approx f(a) + f'(a)\cdot(x-a)$

a should be a "nice" number near x

a) $e^{-0.01}$

from earlier: $e^x \approx 1 + x$ when x close to 0

$x = -0.01$: $e^{-0.01} \approx 1 + (-0.01) = \boxed{.99}$

b) $\sin(91°)$

$f(x) = \sin x$, $x = 91°$, $a = 90° \Rightarrow a = \frac{\pi}{2}$ radians

linear approx. at $a = \frac{\pi}{2}$: $f(x) \approx f(a) + f'(a)\cdot(x-a)$

$f(x) \approx f(\frac{\pi}{2}) + f'(\frac{\pi}{2})\cdot(x - \frac{\pi}{2})$

$f(x) = \sin x \Rightarrow f(\frac{\pi}{2}) = 1$

$f'(x) = \cos x \Rightarrow f'(\frac{\pi}{2}) = 0$

$\Rightarrow f(x) \approx 1 + 0\cdot(x - \frac{\pi}{2})$

$f(x) \approx 1$ (when x is near $\pi/2$)

$\boxed{\sin(91°) \approx 1}$

c) $\sqrt{15.98}$

$f(x) = \sqrt{x}$, $a = 16$, $x = 15.98$

$f(x) \approx f(16) + f'(16)(x-16)$ when x is near 16

$f(16) = \sqrt{16} = 4$

$f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f'(16) = \frac{1}{2 \cdot 4} = \frac{1}{8}$

$\sqrt{x} \approx 4 + \frac{1}{8}(x-16)$ when x is near 16

at $x = 15.98$: $\sqrt{15.98} \approx 4 + \frac{1}{8}(-0.02) = 4 - \frac{2}{800}$

$= 4 - \frac{1}{400} =$

$= 4 - 0.0025$

$= \boxed{3.9975}$

d) $\ln(1.03)$

$\ln(1+x) \approx x$ when x is close to 0

$x = 0.03 \Rightarrow$ $\boxed{\ln(1.03) \approx 0.03}$

e) $(1.01)^{(1.01)} = f(x)\ldots$ Can use $\underline{f(x) = x^x}$

$f(1.01)$, use $a = 1$

$$f(x) \approx f(1) + f'(1)(x-1)$$

$$f(x) = x^x \Rightarrow f(1) = 1^1 = 1$$

$$f'(x) = \underline{x^x(\ln x + 1)} \Rightarrow f'(1) = 1^1 \cdot (\ln 1 + 1)$$

$$= 1^1 \cdot (0+1) = 1$$

$$\boxed{x^x \approx 1 + 1\cdot(x-1) = x}$$ (when x is close to 1)

at $x = 1.01$: $\boxed{1.01^{1.01} \approx 1.01}$

Over or under?

Example: the approximation $e^x \approx 1+x$ near $x=0$

is an underestimate (true value is bigger)

because the graph is above the tangent line

More generally: • if f' is increasing ($f''>0$)

then f grows faster than tangent line $\Rightarrow$

linear approx. is $\boxed{\text{underestimate}}$

• if f' is decreasing ($f''<0$)

then linear approx. is $\boxed{\text{overestimate}}$

Example: is $\ln(1+x) \approx x$ (near $x=0$)

an over or under estimate?

$f(x) = \ln(1+x)$

$f'(x) = \frac{1}{1+x} = (1+x)^{-1}$

$f''(x) = -(1+x)^{-2}$ negative

$\Rightarrow$ overestimate

e.g. $\ln(1.03) = 0.0295\ldots < \text{estimate } 0.03$

Accuracy threshold

Idea: Sometimes we want our approximations to be accurate to within some error threshold.

Example: Find the linear approximation of $\sqrt[3]{1-x}$ at $a=0$ and determine the values of x for which the approximation is accurate to within 0.1

(graphing calculator allowed)

$$L(x) = f(0) + f'(0)\cdot(x-0)$$

$$f(0) = \sqrt[3]{1-0} = 1$$

$$f'(x) = \frac{d}{dx}\left((1-x)^{1/3}\right) = \frac{1}{3}(1-x)^{-2/3}\cdot(-1)$$

$$f'(0) = -\frac{1}{3}\cdot 1 = -\frac{1}{3}$$

$$\Rightarrow \boxed{L(x) = 1 - \frac{1}{3}x}$$

Question: when is $|L(x) - f(x)| < 0.1$?

when $\boxed{-1.2 \leq x \leq 0.7}$

Differentials

Idea: $y = f(x) \Rightarrow \frac{dy}{dx} = f'(x)$ multiply by dx

$$\underbrace{dy = f'(x)\,dx}$$ "differential" of y

★ Note: when calculating differentials, we can leave the expression in terms of x and dx: we don't need to say at what point we are approximating

Example: Calculate the differential of $y = \sqrt{\cos(x)+2}$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{\cos x+2}} \cdot \underbrace{\frac{d}{dx}(\cos(x)+2)}_{-\sin x} = \frac{-\sin x}{2\sqrt{\cos x+2}}$$

$$\boxed{dy = \frac{-\sin x}{2\sqrt{\cos x+2}}\,dx}$$

★ But what does a differential mean?

Idea: we can relate this to linear approximation

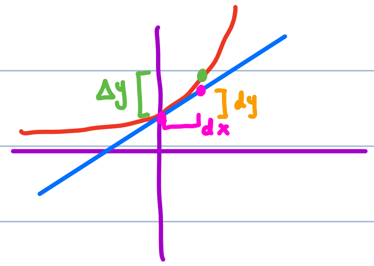
$$f(x) \approx f(a) + f'(a)(x-a)$$

$$\underbrace{f(x)-f(a)}_{\Delta y} \approx \underbrace{f'(a)}_{f'}\ \underbrace{(x-a)}_{\Delta x}$$

Compare with: $dy = f'(x)\,dx$

"dy" tells us the estimated change in y resulting from changing x by dx

Picture:



We can reframe linear approximation in terms of differentials:

$$\Delta y \approx dy$$

Example: Compare the values of Δy and dy if

$y = x^3 - 2x$ and $\underbrace{x \text{ changes from } 2 \text{ to } 2.1}$

Δy = change in y = $y(2.1) - y(2) = \boxed{1.061}$

dy = "predicted change in y" =

$\dfrac{dy}{dx} = 3x^2 - 2 \Rightarrow dy = (3x^2 - 2)\,dx$

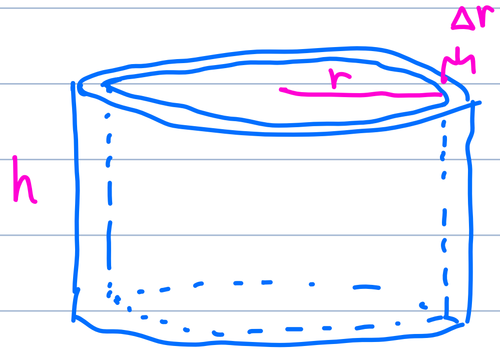
$x=2$ 0.1 (change in x)

$dy = (3 \cdot 2^2 - 2) \cdot 0.1$

$= (12-2) \cdot 0.1 = 10 \cdot 0.1 = \boxed{1}$

<u>Example:</u> a) use differentials to find a formula for the approximate volume of a thin cylindrical shell with height h, inner radius r, and thickness Δr

b) compare to the true value.



← volume = (big cylinder) − (inside cylinder)

(above the two cylinders: volume of, volume of)

$= \Delta V$

$V(r) = (\text{area of base})\cdot\text{height}$

$$\boxed{V(r) = \pi r^2 \cdot h}$$

cylindrical shell = $V(r+\Delta r) - V(r) = \Delta V$

use differentials: $\Delta V \approx dV$

$$dV = 2\pi r h \cdot dr$$

$$\boxed{\Delta V \approx 2\pi r h \cdot \Delta r}$$

b) <u>true</u> volume of cylindrical shell is

$$V(r+\Delta r) - V(r) = \pi (r+\Delta r)^2 \cdot h - \pi r^2 h$$

$$= \pi h\left((\cancel{r^2} + 2r\cdot\Delta r + \Delta r^2) - \cancel{r^2}\right) = \pi h\cdot(2r\Delta r + (\Delta r)^2)$$

Using differentials to approximate error

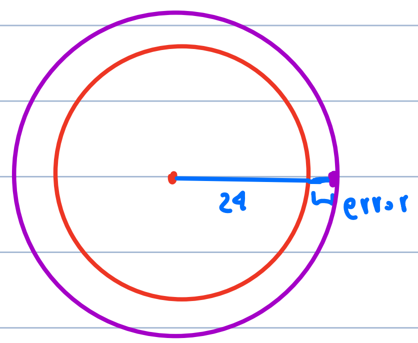
Example: The radius of a circular disk is given as 24cm with a maximum error in measurement of 0.2cm.

error $= |\Delta r|$

a) Use differentials to estimate the maximum error in the calculated area of the disk.

b) What is the relative error?

a)



$r = 24 + \underbrace{\Delta r}_{\text{error}}$

$A = \pi r^2$

$\Delta A = ?$

$dA = 2\pi r \cdot dr$

$r = 24, \quad |dr| < 0.2 \quad \Rightarrow \quad |dA| < 2\pi \cdot 24 \cdot 0.2 = 9.6\pi$

we estimate that potential error in calculated area is at most $\boxed{9.6\pi \text{ cm}^2} \approx 30.16 \text{ cm}^2$

b) relative error $= \dfrac{|\Delta A|}{A} \approx \dfrac{9.6\cancel{\pi}}{\cancel{\pi} \cdot (24)^2} = \dfrac{9.6}{24^2} \approx 0.0167$

% error : $\approx \boxed{1.67\%}$

Example

a) find linearization of $f(x) = (3x+2)^{1/3}$ at $a = 2$

$L(x) = f(a) + f'(a) \cdot (x - a)$

$L(x) = \underline{f(2)} + \underline{f'(2)}\,(x-2)$

$f(2) = (3 \cdot 2 + 2)^{1/3} = 8^{1/3} = 2$

$f'(x) = \frac{1}{3}(3x+2)^{-2/3} \cdot \underbrace{\frac{d}{dx}(3x+2)}_{3} = (3x+2)^{-2/3}$

$f'(2) = (3 \cdot 2 + 2)^{-2/3} = 8^{-2/3} = \frac{1}{8^{2/3}} = \frac{1}{(\sqrt[3]{8})^2} = \frac{1}{4}$

$L(x) = 2 + \frac{1}{4}(x-2)$

b) use this to approximate $f(2.1)$

$f(2.1) \approx L(2.1) = 2 + \frac{1}{4}(2.1 - 2) = 2 + \frac{1}{4}(0.1)$

$= 2 + (0.25) \cdot (0.1)$

$= \boxed{2.025}$